Know More About

How Speakers Work

What Goes Behind That Boom?

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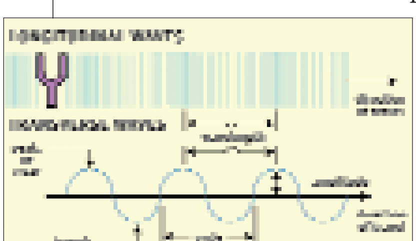
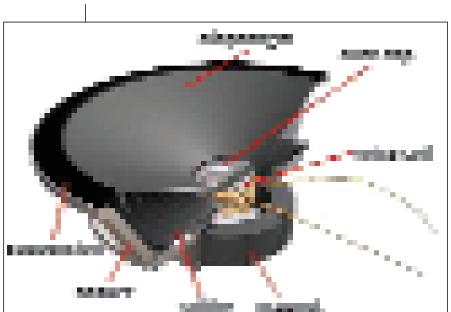
Speakers easily play the most significant role in enhancing your audio experience. So how do they actually manage to reproduce sound from the electrical signals that are fed as input? Devices such as speakers and microphones are technically called *transducers*. A transducer transforms one form of energy into another—in this case, a speaker converts electrical energy into acoustic energy (sound), in about the same way that a microphone converts acoustic energy into a corresponding electrical signal.

There's more to speakers than just plugging them in and believing that the costly sound system you got is going to take you to new heights. Of course, higher-end systems, with their excellent low-noise amplifiers, filters and impedance matchers reproduce crystal-clear sound even from speakers of the lowest end, but audiophiles know the difference!

We need to dwell into some physics to understand its mechanism. During our school days, we all have played around with the tuning fork during our physics practicals. On striking the tuning fork, we hear a long hum.

This sound is created by the compression and rarefaction of air particles, around the vibrating fork arms. When these compressions and rarefactions hit the eardrum, it vibrates, and we perceive sound. In a speaker, this compression and rarefaction is performed by a paper diaphragm that vibrates in accordance with the electrical signal fed to it. The construction of a speaker sheds more light on its functioning. At the base is a powerful ring magnet. In the centre is a voice coil made up of fine wire, wound around a paper cylinder. As we can see in the picture, the voice coil is attached to a spider (an elastic paper sheet that holds the voice coil), such that it always returns to its fixed position under normal conditions.


Sound is produced by the compression and rarefaction of air particles


Sound is created in a speaker by the vibrating diaphragm, similar to the tuning fork

To start off with the functionality of a speaker, the underlying principle is electromagnetism. The basic interaction between an electromagnetic coil and a powerful permanent magnet is what drives the speaker to produce sound. The primary source of sound is a diaphragm. One end of the voice coil is attached to the centre of the paper diaphragm. This diaphragm, in turn, compresses and rarefies the air particles around it, thereby producing sound.

Speaker enclosures, in addition to the speaker itself, also play a vital role in your audio experience. The reason for this is that the air flow through the enclosure drastically affects bass and crackling sounds due to undesirable harmonics. The dimensions of the enclosure have an effect on the resonance of the sound waves.

Getting more into the technicalities, let us consider an amplifier as a block. In this case, it primarily has two signals—the input and the output. For the amplifier to have high gain, its input impedance should be as high as possible and its output impedance should be as low as possible. The reason for this is the Maximum Power Transfer theorem, which states that to get the maximum power output from a source (the amplifier, in this case), the output resistance of the source should be the same, or as close as possible, to the resistance of the load (the speaker, in this case). For those of you who haven't heard of it, this simply means that the resistance of the speaker should be equal to that of the amplifier output. Typical speaker impedances are 4—8 Ohms—you can get these values from the associated brochures and manuals that accompany your sound system.

Finally, in a scenario of multiple speakers, positioning each speaker is of prime importance. Sound, after all, propagates in the form of waves. Although multiple reflections cause sound to be heard across all areas, sound travels in a straight path—like light. You should always consider sound to be similar to light while positioning the speakers around your home, with the prime aim to "illuminate" your living room. This is the easiest way to get it right, with minimal effort.

So go ahead and enjoy those tracks!

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